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DOGR: Towards Versatile Visual Document Grounding and Referring

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Figure 1. We propose DOGR, a multi-modal large language model that enables users to engage in versatile grounded document interactions. The figure illustrates the reasoning results of DOGR for grounding, grounding-and-referring, and referring tasks in DOGR-Bench.

Abstract

*With recent advances in Multimodal Large Language Models (MLLMs), grounding and referring capabilities have gained increasing attention for achieving detailed understanding and flexible user interaction. However, these capabilities still remain underdeveloped in visual document understanding due to the scarcity of fine-grained datasets and comprehensive benchmarks. To fill this gap, we propose the **DO**cument **G**rounding and **R**eferring data engine (**DOGR-Engine**), which generates two types of high-quality fine-grained document data: (1) multi-granular parsing data to improve text localization and recognition, and (2) instruction-tuning data to activate MLLMs' grounding and referring capabilities in dialogue and reasoning. Using the DOGR-Engine, we construct **DOGR-Bench**, a benchmark covering seven grounding and referring tasks across three document types (chart, poster, and PDF document), offering a comprehensive evaluation of fine-grained document understanding. Leveraging the generated data, we further develop **DOGR**, a strong baseline model that excels in text localization and recognition, while precisely grounds and refers to key textual information during conversation and reasoning, thereby advancing document understanding to a finer granularity and enable flexible interaction paradigms. Our code, data, and model are open-sourced at https://github.com/zyinan99/DOGR.*

Work is done during internship at Tencent.

1. Introduction

Visual document understanding is challenging due to dense textual content and complex layouts, which hinder fine-grained comprehension. Effective interaction with such documents requires the ability to refer to specific regions for precise understanding and accurately ground key details.

However, due to the lack of a clear definition for document grounding and referring tasks, as well as the absence of high-quality fine-grained data for training and evaluation, the advancement of multimodal large language models in this field has been notably limited. Many existing models [1, 7, 16, 47] struggle with basic fine-grained document tasks such as text localization and region-level recognition, let alone achieve effective grounding and referring to crucial textual elements during conversation or reasoning.

Several prior works either incorporate multi-granularity parsing data during pre-training to enhance document perception [14] or introduce simple region-level instruction-tuning tasks to achieve basic referring capabilities [24]. While these methods aim to improve document understanding, they all suffer from two significant shortcomings:

- **Suboptimal annotation quality.** To collect parsing annotations, the Optical Character Recognition (OCR) tools are typically employed to extract text and bounding boxes. However, OCR tools encounter issues with inaccurate recognition of text and corresponding bounding boxes and have difficulty extracting semantically coherent text from documents with complex layouts, limiting their potential for developing fine-grained document understanding and question answering data.
- **Lack of diversity in task formats.** Current instruction-tuning datasets primarily cater to basic referring tasks, such as region-level OCR, summarization and translation. However, these datasets fall short in supporting grounding tasks and fail to seamlessly incorporate grounding and referencing capabilities into the dialogue and reasoning processes of MLLMs. The restricted range of tasks hinders the model's ability to flexibly perceive details and affects the user interaction experience.

To solve the above two issues, we introduce the **DO**cument **G**rounding and r**E**ferring data engine (**DOGR-Engine**) for constructing high-quality fine-grained document data. DOGR-Engine generates: (1) 2.1M multi-granular document parsing data, which provides text box annotations at the word, phrase, line, paragraph and full-page level across posters, charts, and PDF documents. This dataset enhances text localization and recognition capabilities and serves as the foundation for creating instruction-tuning data. (2) A diverse set of 700K instruction-tuning samples, covering text-in location-out (grounding), location-in text-out (referring) data, and tasks combining location and text in both input and output. These data, generated from our multi-granular document parsing dataset with GPT-4o [16] assistance, ensure high linguistic quality and accurate grounding and referring annotations.

To benchmark the MLLMs' document grounding and referring capabilities comprehensively, as well as providing a clear task definition,

we propose **DOGR-Bench**, which contains 3.6K test samples and encompasses seven grounding and referring tasks across three document types (chart, poster, PDF document). On this benchmark, the state-of-the-art MLLMs demonstrate poor performance for grounding and referring tasks, suggesting that current models may not yet be equipped to handle the intricate challenges in fine-grained document understanding. This performance gap underscores the DOGR-Bench's value in identifying key limitations and guiding future advancements in MLLMs.

Furthermore, based on data generated by our engine, we develop a strong baseline model, **DOGR**, capable of understanding spatial referring and accurately grounding text within document images. We report the performance of our model on DOGR-Bench, providing a performance reference for future research.

In summary, our contributions are threefold. (1) We introduce DOGR-Engine, a data construction pipeline that generates large-scale, high-quality multi-granular document parsing data and diverse ground-and-refer instruction tuning data. (2) We develop DOGR-Bench, the first comprehensive benchmark designed to evaluate MLLMs' grounding and referring capabilities in document understanding. (3) We present DOGR, a pioneering MLLM that is capable of understanding referred text and performing text grounding during conversation, leading to detailed document comprehension and more intuitive user interaction.

2. Related Work

2.1. MLLMs for Visual Document Understanding

Recently, several Multimodal Large Language Models [6, 12, 22, 54, 59, 66] have been introduced to perform visual document understanding without relying on OCR tools. UReader [59] unifies a wide range of document understanding tasks with instruction-tuning format. A shape-adaptive cropping module is further designed to encode rich textual content in high-resolution image. TextMonkey [29] employs shifted window to build connections between different image tiles, alleviating the issue of incoherence semantic caused by image cropping. InternLM-XComposer-4KHD [12] and InternVL 1.5 [8] further increase the tile number, significantly improving the performance on visual document understanding tasks. These works achieve promising performance but lack document grounding and referring capabilities, which hinders the grounded document understanding and flexible human-AI interaction.

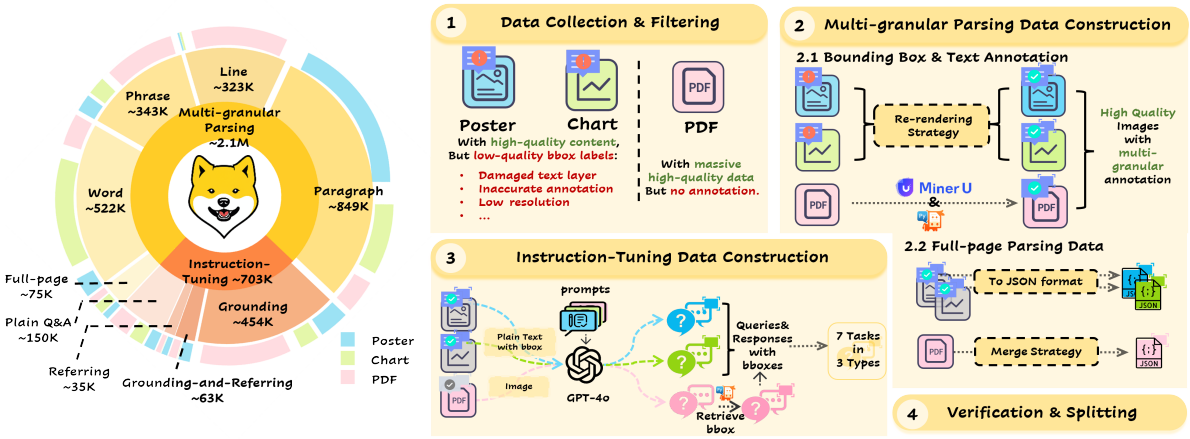


Figure 2. Left: Hierarchical task taxonomy and sample distribution analysis. Right: The pipeline of DOGR-Engine, which outlines the construction process for multi-granularity parsing data and ground-and-refer instruction-tuning data.

2.2. MLLMs for Grounding and Referring

In pursuit of fine-grained image understanding and convenient interaction, recent studies integrate grounding and referring abilities into MLLMs [1, 4, 39, 58, 60, 63]. Kosmos2 [39], Shikra [4] and Ferret [60] utilize bounding boxes or visual prompts to pinpoint specific regions of an image and generate responses with key objects being grounded, facilitating flexible content referring and interaction. Additionally, LLaVA-Grounding [63] and GLaMM [40] further employ finer-grained multi-granularity masks for pixel-level grounding across various semantic levels. These works perform well on real-world images but can't adapt to visual document understanding due to domain gaps. Recently, there are several attempts to develop visual document grounding and referring. mPLUG-1.5 [14] and Kosmos-2.5[30] enhance localization and fine-grained perception with bounding box annotations from OCR tools and PDF parsing tools, respectively. Fox [25] utilizes various visual prompts to refer to document regions and enables the MLLMs to extract or translate the region-level content. However, mPLUG-1.5 and Kosmos-2.5 only support basic text localization and recognition tasks. Fox further supports the referring translation task. These methods, though effective, fall short of integrating both grounding and referring into broader reasonings and dialogues. As a result, they leave the full potential of MLLMs for visual document grounding and referring largely untapped, particularly in reasoning tasks where grounding plays a crucial role.

3. DOGR-Engine

While well-annotated and diverse grounded data are crucial for improving the grounding and referring abilities of MLLMs [23, 39, 49, 62], comprehensive and accurately labeled document-grounded datasets still remain scarce. Given the time-consuming and labor-intensive nature of manually annotating raw document images, we collect a large corpus of documents, including posters, charts, and PDF files, and develop **DOGR-Engine** to efficiently construct fine-grained document-grounded datasets.

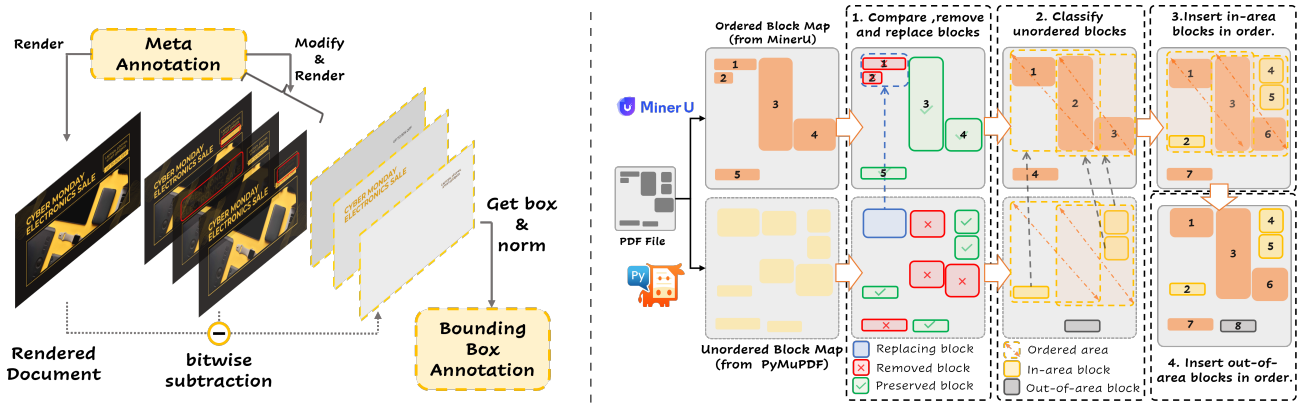
In this section, we first introduce the data processing pipeline and the generation of two types of high-quality document data. Specifically, the data processing pipeline and the annotation process for multi-granular parsing data are described in Sec. 3.1, while the construction pipeline for instruction-tuning data is elaborated in Sec. 3.2. Additionally, an overview of statistical information is provided in Sec. 3.3, and the annotation results are in Appendix A.

3.1. Multi-granular Parsing Data Construction

As shown in Fig. 2(Right), we start by filtering the raw data to remove low-quality samples or those with missing or broken information. We then describe the annotation process for multi-granular parsing data as follows.

3.1.1. Automatic Bounding box Annotation

Poster. We collect poster data from the Crello dataset [57], which consists of poster templates from the design service website. Each poster contains the document meta-annotation, which includes rendered text blocks and their corresponding bounding boxes. However, we observe that some provided rendered text blocks are significantly degraded, and the bounding boxes are inaccurate. To address this issue, we re-render the text content and design a **Re-rendering Strategy** to obtain precise bounding boxes. Specifically, as illustrated in Fig. 3a, the strategy consists of three steps: Firstly, we render the poster with the meta-annotation; Then, we modify the color or opacity attribute of one text block and perform a re-rendering process; Finally, since the rendering results of the first two steps are identical except for the modified attribute, the image of the target text block can be obtained by applying pixel-wise subtraction. By repeating the above steps for every text block, we can obtain accurate bounding boxes of all text



(a) Re-rendering Strategy (b) Merge Strategy

Figure 3. (a) For the poster and chart data, we propose the Re-rendering Strategy to automatically obtain precise bounding boxes. (b) For the PDF document data, we propose the Merge Strategy combining the annotations from MinerU and PyMuPDF to obtain comprehensive and layout-aware full-page parsing annotations.

blocks, which are then normalized to create the final annotation. Note that we preserve the original font format during re-rendering to keep the font diversity of poster data.

Chart. The chart data is collected from the ChartQA dataset [33], including various bar charts, line graphs, and pie charts, along with their corresponding JSON/CSV information. We extract the necessary information for rendering images and utilize Matplotlib to create chart images that closely resemble the original charts. To increase the position diversity of text blocks, we apply random padding around the edges of the chart images. Additionally, to prevent the model from over-relying on text and neglecting its ability to interpret visual elements, e.g., bars and lines, we propose to remove the text data from one-third of the chart data, and randomly mask half of the text in another third. Finally, we reuse the Re-rendering Strategy to obtain bounding box annotations for all text blocks, including the values, titles, legends, and axis labels.

PDF Document. We select a high-quality subset from CC-MAIN-2021-31-PDF-UNTRUNCATED [48], which consists of a large collection of various text documents in PDF format. Leveraging the PDF parsing tool PyMuPDF [11] and the document content extraction framework MinerU [52], we directly extract parsing data of different granularities (word, phrase, line, paragraph).

3.1.2. Full-page Parsing Data Construction

In addition to the detailed annotations across word, phrase, line, and paragraph-level, we also construct full-page parsing annotations to enhance the comprehensive perception of the document content.

Poster and Chart. The poster data features sparse textual content, making it straightforward to grasp the dependencies between text blocks. Therefore, we organize the paragraph-level text and their corresponding bounding boxes in a left-to-right, top-to-bottom scanning order to create the full-page parsing annotation in JSON format. In contrast, given the highly organized nature of the chart data, we structure its full-page parsing annotations as a JSON dictionary containing hierarchically organized components: chart title, axis labels, legends, and data markers—each precisely associated with their spatial coordinates.

PDF Document. PDF document contains extensive text and complex layouts, necessitating parsing in a logical reading order to comprehend the information accurately. MinerU [52] contains a layout detection model and an OCR model, allowing for extracting text blocks with a certain reading order. However, it often fails to capture all content in documents, missing tables, footnotes, and other elements. In contrast, PyMuPDF [11] can thoroughly extract content from documents, but it does not provide an appropriate reading order. Therefore, we propose a **Merge Strategy** that combines the strengths of MinerU and PyMuPDF to achieve comprehensive, layout-aware annotations. As shown in Fig. 3b, 1) we first compare text blocks from ordered and unordered maps to eliminate duplicate text blocks. For the truncated blocks (red block 1 and red block 2 in the ordered map), we replace them with the corresponding complete block (blue block in the unordered map) to improve the semantic completeness within blocks; 2) In the ordered map, we construct an ordered area if two consecutive blocks are placed from the top left to the bottom right. We classify the preserved blocks in the unordered map into two categories: in-area blocks and out-of-area blocks; 3) We insert the in-area blocks into the ordered area and sequentially update the block order in each ordered area using column-major order; 4) For the out-of-area blocks, we insert them into the ordered map. The order of these blocks is then determined based on their positional relationship with their nearest ordered blocks, following column-major order.

3.2. Instruction-Tuning Data Construction

To seamlessly integrate grounding and referring capabilities in the dialogue and reasoning, it is crucial to con-

struct high-quality instruction-tuning data with accurate grounded text in both query and response. Based on the precise bounding box annotations from Sec. 3.1, we leverage GPT-4o [16] to generate diverse-formatted instruction-tuning data tailored to various text granularities, encompassing tasks such as question answering, reasoning, and summarization. To further enhance instruction-following quality, we first constrain the length and type of the generated responses, resulting in different response categories (e.g., short responses, open-ended long responses). Then, we incorporate a response format prompt into each query. The detailed prompts for GPT-4o are shown in Appendix B

Poster and Chart. As introduced in Sec. 3.1.2, the full-page parsing annotation for poster and chart data includes all text blocks along with their corresponding bounding boxes, which contain sufficient information to construct ground-and-refer instruction-tuning data. Therefore, we provide GPT-4o with the full-page parsing data, and task GPT-4o to generate queries and responses in a grounded manner, where the generated content must include texts that originate from the given parsing annotations. Additionally, we require GPT-4o to wrap the texts originating from parsing annotations with '<ocr></ocr>' and append the corresponding coordinates wrapped in '<bbox></bbox>'.

PDF Document. PDF documents often contain extensive text content. If we input all the text of various granularities into GPT-4o, it would result in excessively long prompts. This not only incurs significant API calling costs but also increases the difficulty for GPT-4o in comprehending and completing the tasks, leading to low generation quality. To address these issues, we design a **Post-annotating Strategy** to generate high-quality data with minimal overhead. Specifically, 1) We first use the document image as input for GPT-4o instead of the text of the document, which significantly reduces the token count; 2) Then, we task GPT-4o to generate queries and responses based on the document images. Additionally, any text within the queries and responses that originates from the document image must be wrapped with "<ocr></ocr>"; 3) We extract the texts wrapped with "<ocr></ocr>" and utilize PyMuPDF to retrieve the corresponding bounding boxes; 4) Finally, we wrap these bounding boxes with '<bbox></bbox>' and insert them back into the generated content after their corresponding texts. For texts that cannot be located by PyMuPDF, we remove wrapped tokens and convert them to plain text. This approach allows the generation of diverse instruction-tuning data at a low cost while ensuring the quality of grounding and referring annotations.

3.3. Data Verification and Splitting

Although we require GPT-4o to generate grounded responses in the format of "<ocr> text </ocr> <bbox> x1, y1 ,x2 ,y2 </bbox>", GPT-4o sometimes fails to follow our requirement, resulting in wrong coordinates format or missing "<bbox></bbox>". Therefore, we implement a rule-based filter to remove these defective samples. Additionally, for poster and chart data, we extract the grounded text from the generated content and compare them with the full-page parsing annotations, filtering out samples that contain incorrect grounded text. Furthermore, we also performed a detailed categorization of the data. As shown in Fig. 2(Left), DOGR-Dataset includes 2.1M multi-granular parsing data and 700K diverse-formatted instruction-tuning data across three document types: poster, chart, and PDF document. The multi-granular parsing data includes four fine-grained levels (word, phrase, line, paragraph) and a full-page level, with precise bounding box annotations for the grounded text at different granularities. The instruction-tuning data comprises four types: grounding, referring, grounding-and-referring, and plain Q&A, with the plain Q&A subset derived by removing the grounded content from a portion of the grounding data. This subset is created to enhance data diversity while preserving traditional document understanding capabilities. More dataset statistics and construction details are shown in Appendix D.

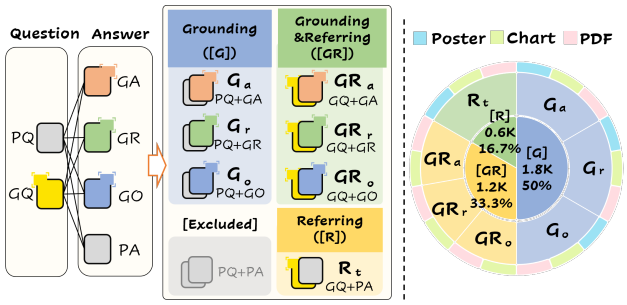


Figure 4. Left: In DOGR-Bench, we categorize the samples into 7 evaluation tasks. Right: Data statistics of DOGR-Bench.

4. DOGR-Bench

To evaluate the grounding and referring capabilities of MLLMs on visual document understanding tasks, we introduce the DOGR-Bench in this section.

Task Definition. As shown in Fig. 4 (Left), we systematically construct our benchmark by categorizing our data into distinct classes based on both input and output formats. This classification helps in designing clear evaluation metrics. We divide the input formats into two categories based on the presence of bounding boxes: **Grounded Question (GQ)** with bounding boxes, and **Plain-Text Question (PQ)** without bounding boxes. The output formats are categorized into four classes:

- **Grounded Answer(GA)**: The response consists of a brief answer accompanied by its corresponding bounding box.
- **Grounded Reasoning(GR)**: The response includes the detailed reasoning process and the final answer, while the

key text contents in the reasoning process are grounded.

- **Grounded Open-ended Answer(GO)**: An open-ended response with one or more key text contents grounded, without providing an answer in a certain format.
- **Plain Text Answer(PA)**: This format does not incorporate grounded text content.

By combining two input forms and four output forms, we derive 7 document grounding and referring tasks. Among these tasks, 3 tasks primarily assess grounding capability: grounded answering for plain text questions (G_a), grounded reasoning for plain-text questions (G_r), and grounded open-ended answering for plain-text questions (G_o). The plain-text answering for grounded questions (R_t) task evaluates referring capability. The remaining tasks, grounded answering with plain-text questions (GR_a), grounded reasoning for grounded questions (GR_r), and grounded open-ended answering for grounded questions (GR_o) require the integration of both grounding and referring capabilities for successful completion.

Metrics. Our benchmark evaluation encompasses two aspects: grounding performance and text answer accuracy. Following the previous works in grounded captioning [60], we evaluate the grounding and text answer separately. For grounding performance, we use $F1_{all}$ score, which evaluates grounding results as a multi-label classification problem. The generated grounded text is considered correct if the Intersection over Union (IoU) between its bounding box and the GT bounding box is greater than 0.5, meanwhile its text matches with the GT text. For text answer accuracy, we use exact text matching accuracy for short-answer tasks and BLEU score for long-answer tasks.

Data Statistics. DOGR-Bench includes 1.8k grounding, 0.6k referring, and 1.2k grounding-and-referring samples. See detailed statistics in Fig. 4 (Right).

5. DOGR

Overall Architecture. As illustrated in Fig. 5, our model DOGR employ a general MLLM architecture, including a vision encoder, a projector, and a large language model. In the visual encoder component, to enable the model to handle high resolution, we first search for the best aspect ratio for the input image and dynamically segment images into multiple tiles. These tiles, along with a thumbnail of the input image, are provided as input to the vision encoder. We employ pixel shuffle[8] to improve the computational efficiency of the model when processing high-resolution images. For bounding boxes, we simply discretize the continuous coordinates into discrete values from 0 to 999, avoiding the introduction of extra modules or location tokens. During inference, we transfer the user-selected regions to coordinates of bounding boxes and insert them into the query for preprocessing. After obtaining the output, we utilize post-processing to overlay bounding boxes on original document

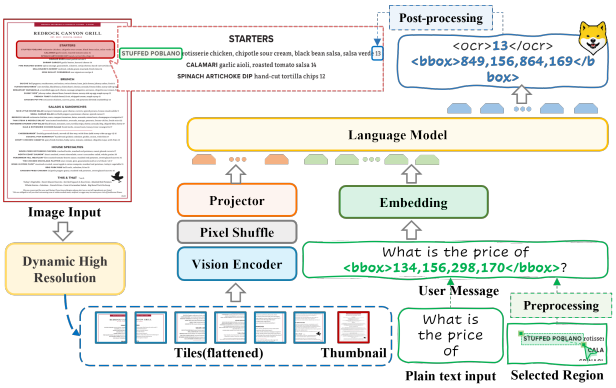


Figure 5. The overall architecture of DOGR.

images, thereby facilitating user interaction.

Training Strategy. We adopt a three-stage training strategy, including pre-aligning, pre-training, and fine-tuning. The pre-aligning stage focuses on aligning the feature space of vision and language rapidly. We freeze both the vision encoder and the large language model, and train the projector with a relatively large learning rate. The pre-training stage aims at document parsing capabilities. We unfreeze the vision encoder and the LLM, enabling the model to recognize diverse textual content and acquire text-reading capability. In the fine-tuning stage, we train the entire model using diverse instruction-tuning data, enhancing its instruction-following ability while activating its grounding and referring capabilities.

Training Dataset. In the pre-aligning stage, we utilize LLaVA-558K [28] to train the projector. For the pre-training dataset, we utilize DocStruct4M [14] along with our 2.1M multi-granular parsing data to enhance basic text reading, text grounding and text referring capabilities of DOGR. For the fine-tuning data, we adopt our ground-and-refer instruction-tuning data and meticulously selected datasets to enhance model performance across a wide range of document parsing and understanding tasks, resulting in a final fine-tuning dataset of 2M samples. The detailed composition of fine-tuning dataset can be found in supplementary materials. In Appendix E, we provide a detailed description and statistics of our training data composition.

6. Experiment

6.1. Implementation Details

DOGR utilizes the InternViT-300M-448px [6, 8] as the vision encoder and Qwen2-7B-Instruct [58] as the LLM. In the pre-aligning stage, we only train the projector and the learning rate is set to 1e-3. In the pre-training and fine-tuning stage, all model parameters are trainable. The learning rate for the vision encoder is 2e-6, while the learning rate for other components is 1e-5. Each stage is conducted for 1 epoch. In the pre-aligning stage, the batch size is set to 256, and we only use the thumbnail for vision encoder input. In the pre-training stage, the batch size is configured

Table 1. Performance comparison on DocLocal4K. † denotes the pre-trained model using parsing data with bounding boxes.

Model	Text Localization					Text Recognition				
	word	phrase	line	paragraph	ALL	word	phrase	line	paragraph	ALL
Gemini 1.5 Pro[47]	6.55	3.69	4.37	5.98	5.18	4.24	2.58	3.41	16.12	11.92
Gemini 2.0 Flash	4.03	5.54	7.36	9.69	6.49	6.18	2.74	3.06	1.01	2.17
Gemini 2.5 Flash[10]	8.57	8.86	10.74	10.72	9.65	14.51	12.45	17.41	37.71	25.04
Gemini 2.5 Pro[10]	23.19	39.11	38.57	10.10	27.91	37.22	32.09	45.39	47.90	46.27
GPT-4o[16]	6.89	2.77	3.18	8.25	5.27	6.02	2.81	4.04	3.86	4.38
InternVL2.5-8B[7]	8.89	5.19	2.60	12.61	7.21	6.88	3.74	4.65	9.08	8.29
Qwen2.5-VL-7B [1]	20.84	13.84	25.25	27.01	21.51	28.68	14.82	22.5	30.51	28.98
DocOwl-1.5†[14]	70.42	76.38	85.88	91.34	80.38	70.10	67.86	73.88	70.70	70.63
DOGR†	**81.85**	**83.58**	**86.88**	**95.67**	**86.64**	**80.00**	**78.85**	**77.60**	**72.72**	**77.88**

Table 2. Comparison of existing MLLMs' and DOGR's performance on DOGR-Bench. For clear comparison, we relax the IoU threshold to 0.1 when evaluating $F1_{all}$(indicated with *). In the final results for DOGR, we set the IoU threshold to 0.5.

Model	Grounding						Grounding-and-Referring						Referring
	G_a		G_r		G_o		GR_a		GR_r		GR_o		R_t
	Acc	$F1_{all}$	Acc	$F1_{all}$	BLEU4	$F1_{all}$	Acc	$F1_{all}$	Acc	$F1_{all}$	BLEU4	$F1_{all}$	Acc
Gemini 1.5 Flash*[47]	32.7	0.7	59.0	1.1	13.6	1.7	42.8	0.5	31.8	0.3	13.5	0.1	37.5
GPT-4o mini*[16]	64.5	0.8	48.7	0.6	9.4	0.5	40.5	0.0	23.3	0.7	7.5	0.2	18.8
Gemini 1.5 Pro*[47]	77.7	9.4	62.0	5.8	13.4	6.7	46.0	5.7	29.8	2.5	17.4	3.7	37.2
Gemini 2.0 Flash*	74.2	5.2	63.5	3.1	15.2	2.9	30.0	0.5	28.5	1.7	26.2	0.6	37.5
Gemini 2.5 Flash*[10]	80.2	38.8	59.3	25.4	12.8	21.5	55.0	22.8	42.0	14.7	16.1	8.6	40.7
Gemini 2.5 Pro*[10]	61.8	30.1	47.8	17.2	13.6	14.7	44.3	17.3	36.0	10.4	19.8	8.7	36.2
GPT-4o*[16]	79.0	8.8	47.0	3.8	12.5	9.2	50.0	0.5	26.8	0.8	12.0	1.2	39.8
Qwen2-VL-7B*[53]	41.7	1.8	4.3	1.4	9.8	5.5	11.8	0.3	0.5	1.5	10.1	0.0	35.5
InternVL2-8B*[8]	51.2	2.4	26.2	0.1	10.2	1.1	26.3	0.0	12.8	0.0	14.5	0.0	17.7
InternVL2.5-8B*[7]	58.3	3.8	33.0	0.6	13.8	2.0	23.3	0.0	15.5	0.0	18.0	0.3	8.5
Qwen2.5-VL-7B*[1]	63.8	19.5	35.0	9.8	6.1	11.3	40.0	1.5	19.0	4.35	8.1	0.6	43.0
DOGR*	83.2	76.3	67.7	54.8	38.3	59.4	82.8	71.4	68.0	39.9	43.0	37.6	60.3
DOGR	**83.2**	**73.0**	**67.7**	**52.5**	**38.3**	**57.1**	**82.8**	**66.9**	**68.0**	**38.2**	**43.0**	**34.9**	**60.3**
- Poster	79.5	75.8	57.0	42.6	45.6	58.9	-	-	-	-	-	-	73.5
- Chart	90.5	72.4	69.5	56.9	32.3	62.8	89.0	74.5	70.0	39.1	45.2	50.7	60.0
- PDF Document	79.5	70.7	76.5	57.8	36.8	49.8	76.5	59.5	66.0	37.3	40.8	19.0	47.5

to 512. The number of image tiles and the max length of the input sequence to the large language model are set to 9 and 4096, respectively. In the fine-tuning stage, the batch size is adjusted to 256, and the image tile number and max input length increase to 16 and 6144, respectively.

6.2. Doc Grounding & Referring Evaluation

Basic capability verification. We first benchmark existing MLLMs on DocLocal4K[14] to evaluate basic text localization and recognition abilities. We report IoU@0.5 for text localization and BLEU-4 score for text recognition at word, phrase, line, and paragraph levels. As quantified in Tab. 1, current SOTA MLLMs[1, 16, 47, 64] show poor performance on both tasks. This underscores the lack of basic capabilities of existing MLLMs in fine-grained document tasks, *i.e.*, detailed text position perception and region-level text recognition. This lack of basic capabilities also limits the model's ability to conduct further flexible grounding and referring. On the contrary, pre-training with document parsing data endows DOGR and DocOwl-1.5[13] with basic capabilities for text localization and recognition. Moreover, our high-quality and rich document parsing data further enhances the performance of DOGR.

Performance comparison on DOGR-Bench. As shown in Tab. 2, our comprehensive evaluation reveals distinct performance patterns among existing MLLMs[1, 7, 8, 16, 47, 53] on DOGR-Bench. Notably, models such as GPT-4o (79.0% Ga Acc) and Gemini 1.5 Pro (77.7% Ga Acc) demonstrate relatively strong performance in basic document understanding tasks, indicating that these models possess fundamental document understanding abilities. However, correspondingly, the lower Acc of referring-related tasks reveals their limitations in accurately perceiving and understanding bounding box coordinates. More crucially, even IoU threshold for $F1_{all}$ is relaxed to 0.1, the other MLLMs' $F1_{all}$ scores across all tasks are consistently low, which indicates deficiencies in precise document grounding. In contrast, our model achieves superior performance in both understanding and grounding capabilities. Most remarkably, its grounding capability demonstrates a substantial advancement, substantially outperforming all existing open-source and proprietary SOTA models across multiple evaluation metrics. The detailed evaluation process and results are shown in Appendix F.

Detailed Performance of DOGR. In the last 4 rows of Tab. 2, we detail the performance of DOGR across 3 data

Table 3. Performance comparison on 10 general document benchmarks.

Model	Size	Doc VQA	Info VQA	Deep Form	KLC	WTQ	Tab Fact	Chart QA	Text VQA	Text Caps	Visual MRC
IXC 2.5-7B[64]	7B	90.9	70.0	**71.2**	-	53.6	**85.2**	82.2	78.2	-	307.5
DocOwl-1.5-Chat[14]	8B	82.2	50.7	68.8	38.7	40.6	80.2	70.2	68.6	131.6	246.4
DocOwl-2[15]	8B	80.7	46.4	66.8	37.5	36.5	78.2	70.0	66.7	131.8	217.4
InternVL2-8B[8]	8B	91.6	74.8	-	-	-	-	83.3	77.4	-	-
Qwen2-VL-7B[53]	8B	**94.5**	**76.5**	-	-	-	-	83.0	**84.3**	-	-
DOGR	8B	91.7	70.7	70.8	**40.4**	**58.8**	84.5	**83.6**	76.6	**145.9**	**332.5**

Table 4. Our multi-granular parsing data MG enhances the model's basic document localization capability.

Model	Text Localization				Text Recognition			
	word	phrase	line	paragraph	word	phrase	line	paragraph
baseline	74.79	78.41	84.1	95.46	79.87	75.73	73.88	70.10
+ MG	**81.85**	**83.58**	**86.88**	**95.67**	**80.00**	**78.85**	**77.60**	**72.72**

Table 5. Our instruction-tuning data IT improves the model's traditional document understanding performance.

Model	Doc VQA	Info VQA	Deep Form	KLC	Chart QA	Visual MRC
baseline	87.57	64.58	66.13	37.10	80.88	265.9
+ IT	**89.24**	**67.45**	**68.89**	**38.32**	**81.92**	**287.25**

types and 7 tasks. Posters have a diverse range of font types and colors, making them ideal for assessing a model's adaptability to different font styles. Charts contain fine-grained and structural elements, necessitating precise text localization and understanding capabilities. PDF document data typically features high-resolution and complex content, posing challenges on both referring and grounding tasks. Despite these challenges, DOGR demonstrates strong performance, validating its effectiveness and robustness. More quantitative results are shown in Appendix C.

6.3. Traditional Document Understanding

To assess the overall capability of DOGR, we conduct experiments on 10 traditional document understanding benchmarks, including DocVQA [34], InfographicVQA [35], DeepForm [44] and KLC [43] for document comprehension, WTQ [38] and TabFact [5] for table understanding, ChartQA [32] for chart comprehension, TextVQA [42] and TextCaps [41] for natural image interpretation, VisualMRC [45] for webpage understanding. For metrics, we use ANLS [2] for DocVQA and InfoVQA, F1 score for DeepForm and KLC, text-matching accuracy for WTQ, TabFact, TextVQA and ChartQA, CIDEr [50] for TextCaps and VisualMRC.

As detailed in Tab. 3, we observe that DOGR achieves competitive performance across all tasks. Notably, DOGR outperforms the previous state-of-the-art model IXC2.5 by +5.3% in accuracy on WTQ and +25.0% in CIDEr on VisualMRC. It is important to highlight that our primary focus is on enhancing the grounding and referencing capabilities, and we do not engage in extensive pre-training and fine-tuning on large datasets as InternVL2 [8] and IXC2.5 [64]. Despite the limited data used for training, DOGR ranks second on three datasets while remaining highly competitive, showing that DOGR maintains a strong performance across general document understanding tasks. We believe that incorporating more data for pre-training and fine-tuning could further improve DOGR's performance on these tasks.

6.4. Ablation Study

Effectiveness of multi-granular parsing data. We evaluate the text recognition and localization performance on DocLocal4K [14]. We set our model pre-trained with only DocStruct4M[14] as our baseline. As shown in Tab. 4, with our multi-granular parsing data, the model achieves performance improvements at all granularities compared to the baseline. Despite the domain gap between our parsing data and DocLocal4K due to the different construction methods, it is evident that incorporating our data improves text grounding and recognition accuracy across various text granularities. This validates the effectiveness of our multi-granular parsing data in enhancing basic text localization and recognition capabilities.

Effectiveness of instruction-tuning data. Besides endowing the model with grounding and referring capabilities, instruction-tuning data further promotes the performance in document understanding. The model fine-tuned using data from DocOwl-1.5[14] serves as the baseline. As shown in Tab. 5, integrating our instruction-tuning data with DocOwl-1.5's data leads to a 2.87% increase in ANLS on InfoVQA and a 21.35% increase in CIDEr on VisualMRC.

7. Conclusion

In this paper, we introduce DOGR-Engine, a data construction pipeline for generating high-quality ground-and-refer data for fine-grained document understanding. Additionally, we construct DOGR-Bench as the first comprehensive benchmark for evaluating the document grounding and referring capabilities of MLLMs. Furthermore, leveraging data generated by our engine, we develop DOGR, a pioneering MLLM that integrates text grounding and referring abilities into the dialogue and reasoning process. Our results show that DOGR can achieve versatile document grounding and referring while achieving promising performance on traditional document understanding tasks. We hope our work will facilitate practical document AI assistants.

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